

Analyzing the impact of the form of navigational guidance on spatial working memory in three-dimensional environments

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While the topic of spatial memory, often interlinked with spatial navigation, in virtual environments has been studied extensively, little research has been done in trying to determine if a specific method of navigation outperforms other methods in terms of helping a person remember the virtual space they traversed. In this paper, we conducted an experiment using the videogame Minecraft with 36 participants, having them collect multiple items in multiple mazes using the following three navigational methods: (1) following written instructions, (2) following signs, and (3) using beacons. Our results did not show a significant improvement in spatial memory depending on the navigation method used, except that the usage of beacons aided participants in memorizing the specific location of an item better than the usage of signs. These underwhelming results are most likely due to a subpar set of survey questions, an imbalance in map difficulty, and a small sample size.

1. Introduction

Three-dimensional environments offer a special framework for finding out how people grasp and use spatial data. Be it in virtual scenarios, real-life structures, or engaging digital universes, individuals are incessantly dependent on their working memory to understand the environment around them and to move in a correct way. However, today's real and digital environments are also frequently equipped with various means of navigation that influence this way of working. These external aids may assist users in finding their way, yet they might also reduce the cognitive engagement required to construct a robust spatial mental model.

This project explores that tension. By placing participants in controlled 3D environments and varying the form of navigational guidance they receive, we aim to understand how different types of cues might influence spatial working memory. The underlying idea is straightforward: some navigation styles may promote active processing of the environment, while others could offload the demand onto external prompts. This study therefore

investigates how guidance strategies affect a participant's ability to recall and reason about the spatial layout they have just explored.

1.1 Background

Working memory is one of the major factors that determine the success of spatial navigation. People who tend to get lost in an environment are the ones who do not constantly store and update information about landmarks, paths, and spatial relations. Highly explicit or directive guidance may facilitate immediate task performance but weaken long-term spatial understanding. Conversely, minimal or indirect cues may require deeper engagement with the environment, thereby strengthening internal representations.

Virtual environments have become increasingly valuable for studying these dynamics. They offer the possibility to create controlled yet realistic 3D spaces where the presence, type, and intensity of navigational guidance can be systematically manipulated. Unlike traditional laboratory tasks, virtual worlds allow participants to explore spaces that behave consistently while still retaining ecological complexity. This makes them ideal platforms for examining how different forms

of assistance affect spatial learning and memory formation.

Within this growing body of work, relatively little attention has been given to how specific forms of navigation cues—textual vs. symbolic vs. highly salient visual indicators—affect the development of spatial working memory. Because these cue types differ in how much they direct attention or reduce cognitive demands, understanding their comparative effects can shed light on broader questions about how people internalize 3D spaces.

1.2 Problem Definition / Research Question

Although navigation support systems are now ubiquitous ranging from written instructions to visual markers and digital path indicators the cognitive impact of these tools remains incompletely understood. Navigation aids can guide users efficiently through a space, but they may also reduce the need for individuals to construct detailed internal spatial representations. This raises an important question for both cognitive science and design practices: does the type of guidance provided alter how people remember and mentally organize a three-dimensional environment?

With this in mind, the study focuses on the following research question:

To what extent does the form of navigational guidance shape spatial working memory in three-dimensional environments?

The question allows us to compare how varying degrees of explicitness and visual salience influence a participant's ability to recall spatial layouts after exploration. It also enables us to examine whether some guidance styles support deeper spatial learning while others primarily promote surface-level task performance.

1.3 Relevance

Addressing this research question is valuable across multiple domains. First, in game design and virtual learning environments, designers frequently choose how much navigational support to embed into their worlds. If different forms of guidance lead to a decrease in spatial learning, then these options have consequences for the educational or experiential goals of the environment. Knowing the cognitive trade-offs can be instrumental in deciding when highly explicit cues are helpful and when they can be detrimental.

Secondly, the research results have a bearing on the use of navigation technologies in the real world. With the advent of GPS applications, augmented-reality directions, and in-building guidance systems usage, which are becoming increasingly popular, concerns have been raised that excessive reliance on such tools may lead to the loss of spatial memory. The experimental data from 3D environments may shed light on whether certain types of guidance that support externalization of cognitive processes are the ones that usually happen internally.

Lastly, this research is a reminder of the methodological potential of using game environments as tools for investigation. The virtual worlds offer a great deal of control over the change of navigational cues while at the same time being able to provide a realistic feeling of movement and spatial immersion. By proving that such environments can effectively assess the differences in spatial working memory, the authors of the study are advocating for their wider application in the domains of behavioural and cognitive research.

1.4 Minecraft as an Experimental Platform

When designing the experiment, an early consideration involved selecting the most suitable platform for building and controlling the 3D environments. We initially explored several alternatives, including developing a custom prototype in the Godot game engine or constructing the experiment in Garry's Mod. Both options offered flexibility, yet they also introduced practical constraints. Creating a custom environment would require significant development time, while tools like Garry's Mod come with their own learning curves and potential mod-related inconsistencies.

In the end, Minecraft was the best fit for our goals. We were able to build each map very accurately with the help of its creative tools, thus all participants faced the same spatial layouts and navigational cues. This aspect was of great importance as our experiment was based on the use of three different types of guidance i.e. written instructions, signs, and beacons, which had to be implemented in a consistent and controlled manner. Using Minecraft it was very easy to embed these cues to the surroundings and at the same time, the remaining parts of the spatial structure could be kept identical for different conditions.

Additionally, Minecraft's environment is straightforward to manipulate, enabling rapid iteration and making it feasible to produce multiple map variations without building a full game from scratch. Since each layout had to be recreated under all three navigation conditions, this flexibility significantly reduced development time and ensured uniformity between versions.

A further advantage is familiarity. All members of the research team possessed prior experience with the game, which streamlined the development process and reduced the overhead associated with environment design. Participants also generally understood its basic controls with minimal instruction, ensuring

that performance differences were not confounded by lack of gaming experience.

In essence, Minecraft offered a platform that was both adaptable and immediately usable. It maintained the ecological richness needed for studying real navigational behaviour while providing enough experimental control to systematically vary the form of guidance presented to participants. Its accessibility, flexibility, and ease of use made it an effective foundation for exploring how different navigation styles, namely written instructions, signs, and beacons, shape spatial working memory in a three-dimensional environment.

2. Previous Research

Research on spatial memory is often interlinked with how people navigate various spaces, be they 3D virtual reality environments (Foo et al., 2005) or flat 2D mappings on a computer screen (Kerster et al., 2016). The research is prevalent in the fields of neuroscience and medicine, where the focus is commonly directed towards what parts of the brain concern spatial memory (Sullivan et al., 2024) and how spatial memory and navigation can be of use in maintaining and improving a person's mental fitness (Merriman et al., 2022).

Furthermore, while not relevant to this study's aims, the wealth of research on spatial memorization frameworks and strategies (allocentric and egocentric) are also worth mentioning (Fernandez-Baizan et al., 2019; Nett et al., 2025). These studies are also often positioned in the same topics as those mentioned above, in the hope to understand how spatial memory and navigation can be impacted by brain diseases (Cogné et al., 2018).

There has been a lack of research focusing on how certain methods of navigation impact a

person's spatial (working) memory of a three-dimensional environment and whether a specific navigation method helps people remember a space better. Due to this, the focus of this section will be on previous research concerning two distinct topics in the field of memory (as a whole) and navigation which were vital to how we shaped our experiment. How these topics impacted the design choices of our experiment is explained in more detail in the methodology section of this paper. The topics are (1) the impact of text expectation on memory, and (2) the impact of landmarks on spatial memory.

2.1 Text Expectation

Any research regarding the memory of a participant needs to decide whether the participant is allowed to know that they are tested on their memory or not. Previous research has indicated that this expectation or lack thereof does impact a person's memory. Wamsley et al. (2016) showed that, if participants were aware that they would be tested on their memory, their spatial and motor memory was enhanced compared to those who were not aware of any test. This result is partly supported by research done by Oberauer and Greve (2022), who indicated that people who intentionally remember information are better at recalling that information if it was stored in their working memory. However, they also provide evidence that intention to remember information does not have a significant impact on a person's (episodic) long-term memory.

2.2 Landmarks

A recurring theme in the field spatial memory and navigation is the importance and significance of landmarks. Research has consistently shown that landmarks, which we Werkhoven et al. (2014) defines as "typically distinctive objects that stand out in the environment and serve as reference points when we are following routes or when we need to determine where we are", are capable

of enhancing a person's spatial memory and navigation (Merriman et al., 2018; Ruotolo et al., 2019). Additionally, adding an audible element to the landmark further reinforces this positive impact (Werkhoven et al., 2014). Furthermore, Wang et al. (2025) has shown that, in addition to landmarks, the shape and color of the boundaries of the space a person navigates (e.g., the walls) also influences their spatial memory in a positive manner, even more so than the landmarks that inhibit the space.

Finally, a distinction should be made between local landmarks (only visible from certain locations relatively close to the viewer) and global landmarks (visible from any position in the environment). According to research done by Credé et al. (2020), participants were better able to remember the layout of the virtual environment they walked through if global landmarks were present, rather than local landmarks.

3. Methodology

3.1 Navigation Methods

Since the objective of our study was to compare working memory across different navigation methods, we decided that testing three different methods of navigation would be the best option, because testing two different methods meant that we would have to use a control group to compare our data with, and four or more methods meant an increase in the minimum needed participants. This would be outside the scope of this project's time and resources.

When deciding on the types of navigation methods that we wanted to incorporate, we went for methods that are commonly present in 3D games that would be easy to reproduce in a Minecraft maze: (1) written instructions in a book that could be viewed by the players, (2)

signs pointing to the correct path, and (3) beacons that are visible from a distance. The written instructions could be found in a book that would spawn in the player's possession when they would enter the map. They could open the book at any time, but this did mean that they were not able to move during this time or look at both the instruction and the map at the same time. The signs were placed on every intersection. This was necessary to prevent players from making wrong turns. Every sign would contain the name of the item and an arrow pointing to the correct path. The beacons were placed two blocks away from the items. This meant that players were able to locate the items by moving towards the beacons and deciding which path would take them there.

3.2 Map Design

To get to the map design we made a number of sketches that we thought would be challenging to navigate, but not too big or complex to remember so that it would be too much for their working memory to store. We selected the one that matched this criterion the best and we used it as a reference when designing the three maps. This allowed us to come up with three map designs that were similar in difficulty and size. The item placement within the maps was included in the sketches. We added landmarks to each map so that participants would have some points of interest that they could memorise or that could help them navigate the maps. These landmarks were carefully chosen and compared between each map to make sure that no map had a combination or number of landmarks that would make that map easier to remember. To realize these designs we used the WorldEdit plugin that made it easy to construct large floors and walls. It also made it possible to copy and paste maps to be used for a different navigation method. The items were placed in chests which the participant could see and open. Images of the maps and

navigational methods as they appeared in the game can be found in Appendix A.

3.3 Experiment

41 participants were tested, 33 male and 8 female ranging from age 20 to age 33. Participants were recruited from personal social circles with no specific requirements. 1 participant had to stop during the experiment due to motion sickness. The survey data of 4 participants were not incorporated in the results because of early flaws in the experiment and changes made to the maps. All participants had to sign a consent form before conducting the experiment. Results were collected via a Qualtrics online survey. The experiment could be conducted both in person with a single laptop or PC present, as well as online using a dedicated Minecraft server where the supervisor would be using a computer of their own to monitor the participant. In both cases the role of the supervisor would be the same: explaining the objective of locating and collecting three items in each map, making sure the participant was familiar with the game mechanics, explaining the navigation types, and timing the participant in all three conditions. Participants were not told that they were being tested on their memory. It was only after the experiment that participants were told that they had to fill in a survey about the experiment. We made a control room where the participant would spawn in, to make it easy to explain the objective and game mechanics. From here the participant could be teleported to each map using a set of in-game buttons in the control room. Once in the map, the participants would immediately start the search for the items. After an item was located and transferred to the inventory of the player, an in-game message would appear that told them which item they collected. When the players collected the last item, they would be killed by the supervisor, which would result in them spawning back in the control room. A more detailed description of the inner workings of

the game world (e.g., teleportation, welcome messages) can be found in Appendix B.

3.4 Permutations

Working memory fades with time, and since our participants had to complete three maps that could each take up to 5 minutes to complete, there could be a difference in the memorization of the first map or navigation method played compared to the last one. To control for this outcome, we made sure that both the maps and the navigation types were played in every possible order. This meant that there were 36 map-navigation permutations to be played, with each participant having played one of those permutations.

4. Results

Data was collected from 36 people in total; each participant used all three navigational variants (beacons, written instructions, and signs) and explored each map in a different order, and filled in a questionnaire after the experiment. In this section, the most important and compelling results of the questionnaire (and in-game metrics) are shown.

4.1 Time Spent on the Maps

Time spent on the maps

The time players spent on each map was recorded; this was not mentioned to participants during testing to avoid influencing their performance. The results are grouped by navigation method, and the averages are shown below (in minutes: seconds):

Written instructions:	3:10
Beacons:	2:10
Signs:	1:49

4.2 Map Order

This question was stated as follows: “Please rank the colors of the maps in the order that you played them”. The results of this question were cross-examined with the permutation of the participant to determine what navigational

method they used and to compare their “correct” order to their response. Their accuracy in answering this question was determined by attributing an “accuracy” of 33.333...% for each color they put in the correct place. When examining these results over all participants, it was found that the type of navigation method used first did not matter. The navigational method did not affect the participants' recall of the map order. As shown in Figure 1, the accuracy percentages are nearly identical across the methods.

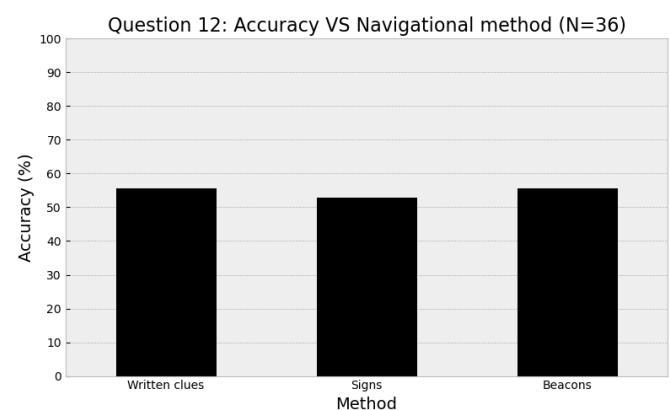


Figure 1: Results of the accuracy of responses to question 12 compared to the navigational method used for the color map.

4.3 Remembering Map Layout

Question 13 was as follows: "Looking at the sketches [in Figure 2], which one best matches the white map you explored?". The correct answer was the left map.

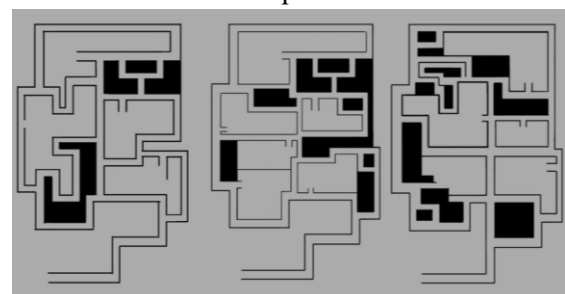


Figure 2: Maps where the participants could choose which one they thought was the white map, with the left one being the correct map.

As shown in Figure 3, 72% selected the incorrect map. This result is close to what is expected when people are guessing randomly,

which would be 2/3 or 67% incorrect. However, upon closer examination of the participants' responses (shown in Figure 4), we observed that participants answered more correctly the later the white map was in their permutations. We also observed that participants who used written instructions on the white map answered correctly more often than those who had beacons or signs to guide them. For the latter result, we used Fisher's exact test for pairwise comparisons, which indicated that the difference was not statistically significant ($p = 0.965$).

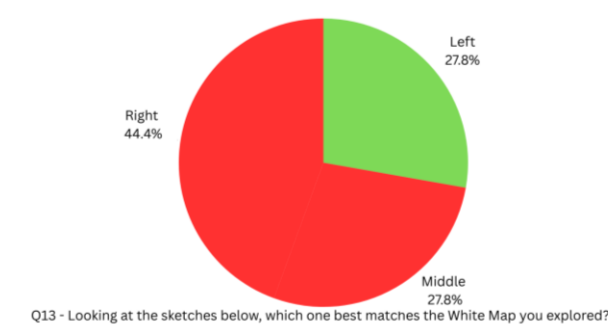


Figure 3: The overall responses to Question 13. The correct answer was "Left", which was answered by 27.8% of the participants. This question had a sample size of $N=36$.

4.4 Map Complexity

Question 16 is "Which color map do you think had the most complex layout?". This question

was asked because we suspected that participants would spend more time on a more complex map, regardless of the navigational method they used. The results (shown in Figure 5) indicate that the red map was the most complex, with 18 votes; the blue map was ranked second, with 10; and the white map ranked last, with 8. Alongside question 16, question 17 asks the participants: "Which color map do you think had the simplest layout?" The results indicate that participants rated the white map layout as the simplest (15 votes), followed by blue (11 votes) and white (10 votes).

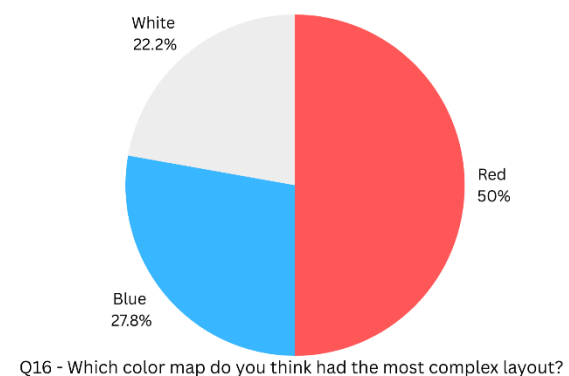


Figure 5: The responses to Questions 16 (top) and 17 (bottom). Red was voted to have the most complex map layout, followed by blue and then white. White was selected as having the simplest map layout, followed by blue and red.

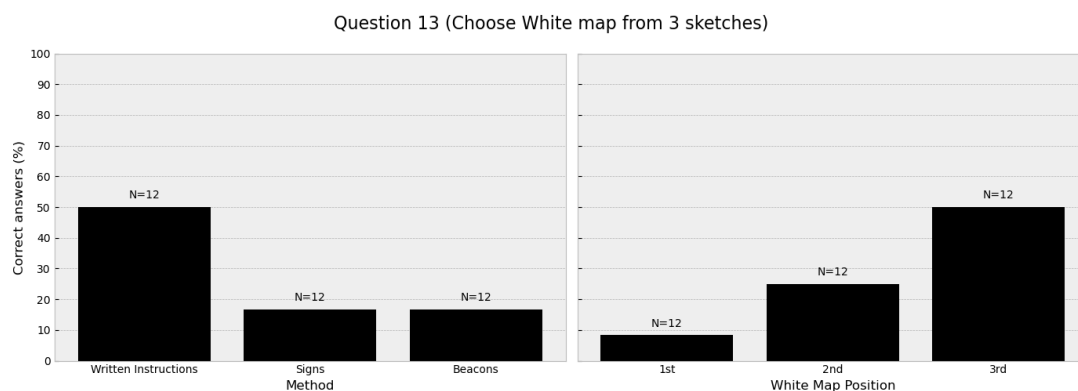


Figure 4: This shows the fraction of participants that answered question 13 correctly. The left panel groups participants by what method they had on the white map and the right panel groups them by what position they played the white map in their permutation.

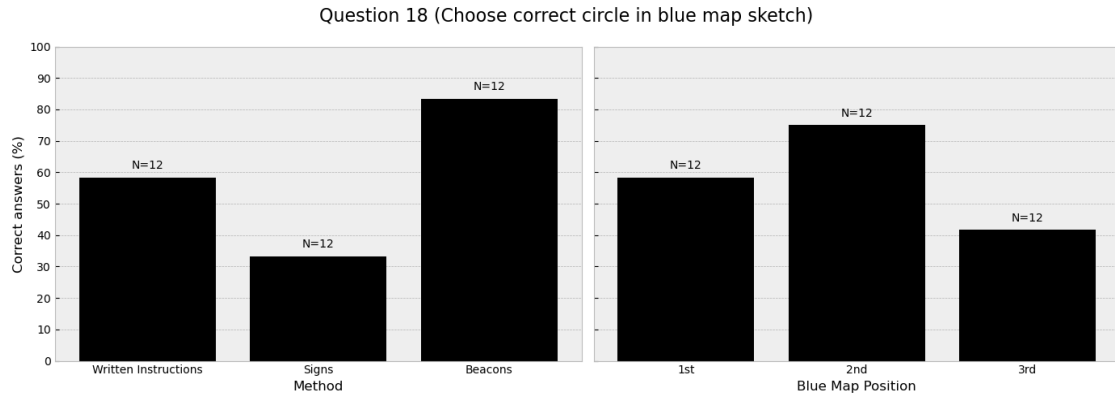


Figure 7: This shows the fraction of participants that answered question 18 correctly. The left panel groups participants by what method they had on the blue map and the right panel groups them by what position they played the blue map in their permutation.

4.5 Remembering Item Location

For question 18, we asked the participants to select the correct option regarding the location of an item on the blue map (Figure 6). Option 1 was the correct location. 61% of the participants were correct. When examining the results in greater depth (Figure 7), it is evident that participants who played the map with beacons remembered the locations of the items better than in the other two navigational variants. The order in which the participants played the maps did not influence the results. The only significant result we obtained was that the beacons' score was higher than the signs' score ($p = 0.0180$).

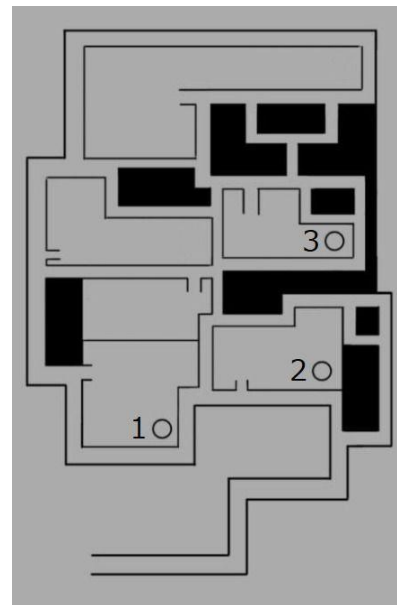
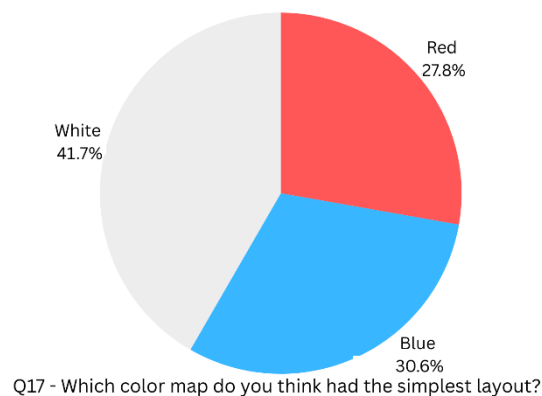


Figure 6: A sketch of the blue map with 3 marked locations, which is used for question 18.



4.6 Impact of Landmarks

For this question, we asked the participants how much the landmarks helped them memorise the map layout. As shown in Figure 8, 21 participants reported that the landmarks were not helpful.

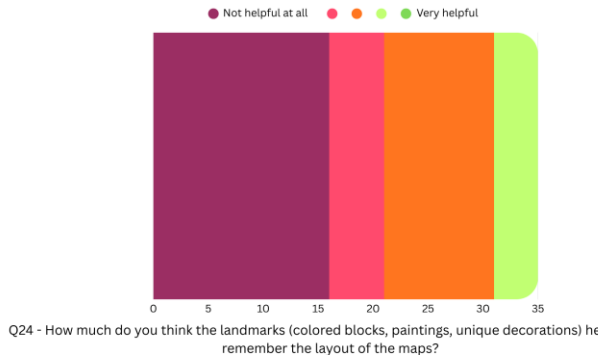


Figure 8: Answers from participants when asked if the landmarks helped them remember the layouts.

5. Discussion

There is a wide variety of results that were obtained from this research, which will be discussed in this section in a rough order of significance and importance.

5.1 Memory of the Map Layout

Question 13 tests how well a participant had memorised the overall layout of the maps and how effectively they could translate this memory into a 2D mapped projection and differentiate between the sketched maps. However, a participant may not have needed to memorise the entire layout of the map, as recognising certain corners, intersections, or other distinctive features of the map could also lead them to the correct sketch. Overall, this question appeared to be difficult. As shown in Figure 3, only 27.8% of participants answered correctly, close to the 33.3% accuracy that is expected from randomly guessing one of the three options. When comparing the accuracy across the navigational method, participants using the written instructions seem to perform better. However, this difference was not statistically significant ($p = 0.965$), indicating that there is no evidence that any of the navigational methods meaningfully impacted performance on this question. This could be explained by the relatively small sample size of the study.

5.2 Memory of the Item Locations

Question 13 and 18 might seem similar, but they test the working spatial memory a slightly different way. Question 18 tests how well a participant can recall where the specific items were in the map and where the navigational aids led them to, rather than testing the memory of the whole layout and testing it against other sketches.

The results of this question in Figure 7 are different to those of question 13. Here, the participants who had beacons as a navigational aid on the blue map answered correctly most often (83%), followed by written instructions (58%) and then signs (33%). The position that they played the blue map in also shows no correlation like it did for question 13. When comparing the results grouped by navigational aids and testing the significance, only between the beacons and signs is the difference significant ($p = 0.018$), whereas the difference between beacons and written instructions ($p = 0.185$) and written instructions and signs ($p = 0.207$) are not significant. This means that the only statement that can be truly made is that participants that had beacons as a navigational aid on the blue map answered correctly significantly more often than participants who had signs.

A proposed explanation for this is as follows. Beacons act as a global landmark, which can indicate to the participant at all times exactly where each item is, whereas the signs only lead the participants to an item once and are then ignored. This makes it so the participant is shown much more often where the items are and the participants therefore remember their locations better. Furthermore, because multiple beacons can be visible at the same time depending on the location of the participant, the relative distance between each item is also clearer compared to having signs pointing in the general directions of the items.

5.3 Time Spent on the Maps

Another factor that could have influenced the results of questions 13 and 18 are the time that the participants spent on average in a map with each of the navigational aids.

The participants spent on average the least time in maps when using signs as navigational aids. On both questions, signs also scored the (joint) lowest. A participant might remember less about a map when they have spent less time in it, meaning that this could have influenced the results. However, there is not enough data or evidence to say how big this influence could be.

The participants spent on average the most amount of time in maps where they had written instructions as a navigational aid. This is likely also hugely impacted by the fact that players had to stop and read the instructions from the book they carried with them. While reading these instructions, the rest of the screen and thus the map was obscured by the book. This time could not be spent navigating, exploring, and subconsciously memorising the map, like with the other methods, causing the total time spent in the map to be inflated compared to the actual time navigating, exploring and subconsciously memorising the map. Therefore, we cannot make any claims about the impact of this difference in time spent on the maps between then written instructions and the other navigational methods.

5.4 Memory of the Map Order

The accuracy of the responses to question 12 were very similar when grouped by navigational method, ranging from 52.7% to 55.6%. This indicates that the navigational method had no clear effect on participants' performance for this question. However, question 12 does not actually test spatial working memory, rather it simply asks participants to recall the order in which they played the maps, without requiring them to remember any spatial features or layouts.

Because our research is specifically focused on the impact of navigational methods on spatial working memory, the lack of a clear correlation in the results of this question will not meaningfully affect the conclusions of this research.

5.5 Map Complexity

A goal in designing the three different maps was to make sure that they all had a similar difficulty and complexity. To evaluate if this was achieved, participants were asked in questions 16 and 17 to choose the maps with the most complex and simplest layout respectively. These responses are analysed in two ways. First, the full distribution of responses is compared to an equal, random distribution of 33%-33%-33% using a chi-squared goodness-of-fit test. Second, it is tested if the most frequently chosen map in each question was chosen significantly more often than expected by random chance using a one-sided binomial test.

For question 16, the chi-squared test did not indicate a significant deviation from an equal distribution ($p = 0.097$), but the binomial test showed that the red map was chosen as having the most complex layout significantly more often than expected by random chance ($p = 0.028$). For question 17, the chi-squared test was not significant ($p = 0.558$) and white was not chosen significantly more than by random chance ($p = 0.19$).

From this we can conclude that, although the different color maps were perceived as roughly comparable in complexity, the red map was still significantly more complex than the others. This suggests that the design goal of having the maps be of similar complexity was only partially achieved.

5.6 Landmarks in the Maps

As mentioned in Section 2.2, landmarks were added to each map in an attempt to improve a participant's ability to remember the layout of the map. To evaluate if this was done

successfully, we asked participants in question 24 how much they thought the landmarks helped them remember the layout of the map on a 5-point scale from “Not helpful at all” to “Very helpful”. 21 participants or 58% reported that the landmarks were not helpful. Including the central point as neutral, that figure goes to 31 participants or 86% who did not report the landmarks as useful in remembering the layout of the map. This suggests that the landmarks had no perceived added benefit to the participants’ performance in memorising the maps.

Two things should be noted, however. Firstly, this result is not entirely reliable, as it relied on only the perceived helpfulness, and it was not tested itself. This means that it is still inconclusive if adding landmarks was a good choice in the map design that was beneficial to the experiment.

Secondly, landmarks used in previous research were often abstract items that did not fit in with the rest of the environment (Werkhoven et al., 2014; Wang et al., 2025). This may have made the landmarks stand out and be noticed more, which may have impacted the improvement in memory. The landmarks that were put in the maps in this experiment looked more like decorations that were not very abstract and impact on memorisation performance may therefore have been weaker or even negligible.

5.7 Limitations

There are four main points that limited this study and made the results less conclusive than was hoped.

5.7.1 Sample Size

This study ran the experiment on a total of 36 participants. While this was enough to make sure that each permutation was played at least once, it turned out to not be quite large enough to draw strong conclusions about subtle differences or statistically significant effects. Although getting 36 participants to do the experiment within the limited timeframe was

already quite a feat for this group, having a larger sample size with multiple participants per permutation would have increased the reliability of the results.

5.7.2 Questionnaire Design

A second limitation lies in the design of the questionnaire. In retrospect, some of the questions we asked the participants did not contribute meaningfully towards answering the main research question. Some also relied on self-reported experiences, which did not lead to meaningful results. A more effective questionnaire would have focused more on directly testing the spatial working memory of the participant, rather than subjective evaluations.

5.7.3 Map Complexity

As mentioned before in Section 5.5, the three different maps were not completely similar in the complexity of their layouts, as the red map was judged to be significantly more complex than the other two. However, as every permutation varied the navigational type on each map, as well as the order of the maps, this might not have had a significant impact on the results. Nevertheless, we think future work could benefit from testing and confirming beforehand that the maps are of similar complexity, before running the experiments.

5.7.4 Participants’ Surprise About the Map Colors

We noticed that when participants had finished playing the maps and started answering the questionnaire, a lot of the participants were surprised about the fact that the three maps they played had a different colored floor. They were not told this information before playing. We think that this caused them to guess randomly more often than if they knew this. Therefore we think that it would have been beneficial to add this piece of information to the participants before they played the maps.

6. Conclusion

In this section, we briefly revisit the research question for this project: "To what extent does the form of navigational guidance shape spatial working memory in three-dimensional environments?" To gather information on this question, we conducted an experiment in Minecraft in which we asked our 36 participants to find three sets of items across three differently colored maps, with each time the aid of one of three navigation methods we had chosen: a book with directions towards the items, signs on the walls directed towards the items and a beacon next to each item. During the experiment, without informing participants, we timed how long it took them to complete all items on the three maps. After conducting experiments in Minecraft, they needed to complete a survey about the experiment.

The overall results from the experiments were insignificant. People believed that the landmarks did not improve their ability to memorise their environments.

The issues that arose during this project were: the maps were not equal; the red map was perceived as more difficult than the others, with 50% of participants selecting it as the most complex. During the first three tests, the survey and the maps had a few errors. Those results were not used for this report. In the survey, the biggest mistake was not having enough questions about whether participants had memorised the maps and including too many irrelevant questions.

The limitations this project had were: The questions were not always practical; alternative questions could have been asked. The sample size was too small, even though getting 36 participants to test was already a difficult feat for our group. As a result, all permutations were played only once, allowing multiple plays could have yielded more and higher-quality data for comparison. In light of these limitations, future work would involve

recruiting additional participants and incorporating more research-relevant questions into the survey.

To summarize, comprehensiveness and validity of future studies could be enhanced by addressing the limitations faced by this study. This should provide a better understanding of the impact of different navigation methods on a person's spatial memory.

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Appendix A

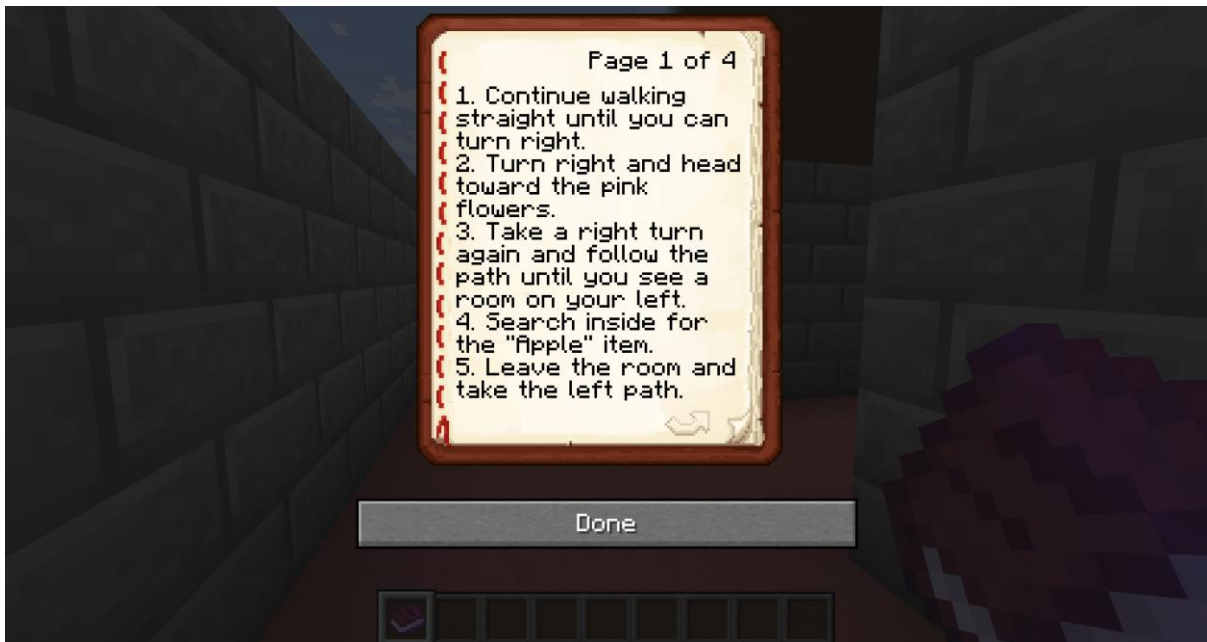


Figure 1: An example of some of the written instructions participants had to follow



Figure 2: An example of some of the signs participants had to follow.

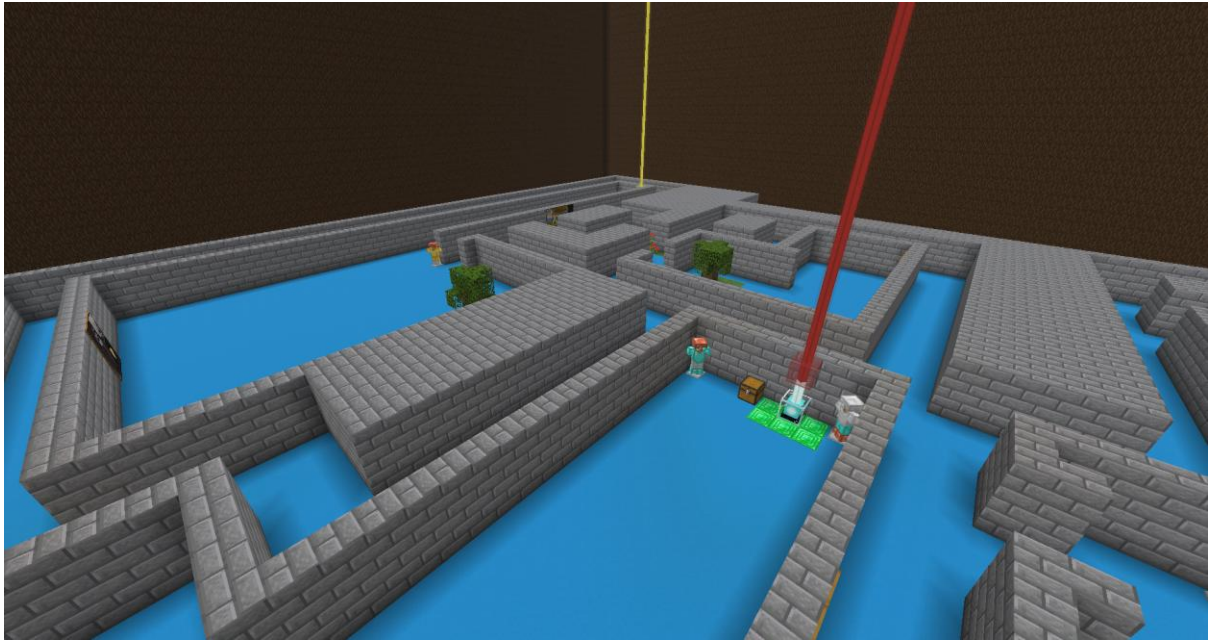


Figure 3: An example of some of the beacons participants had to follow.

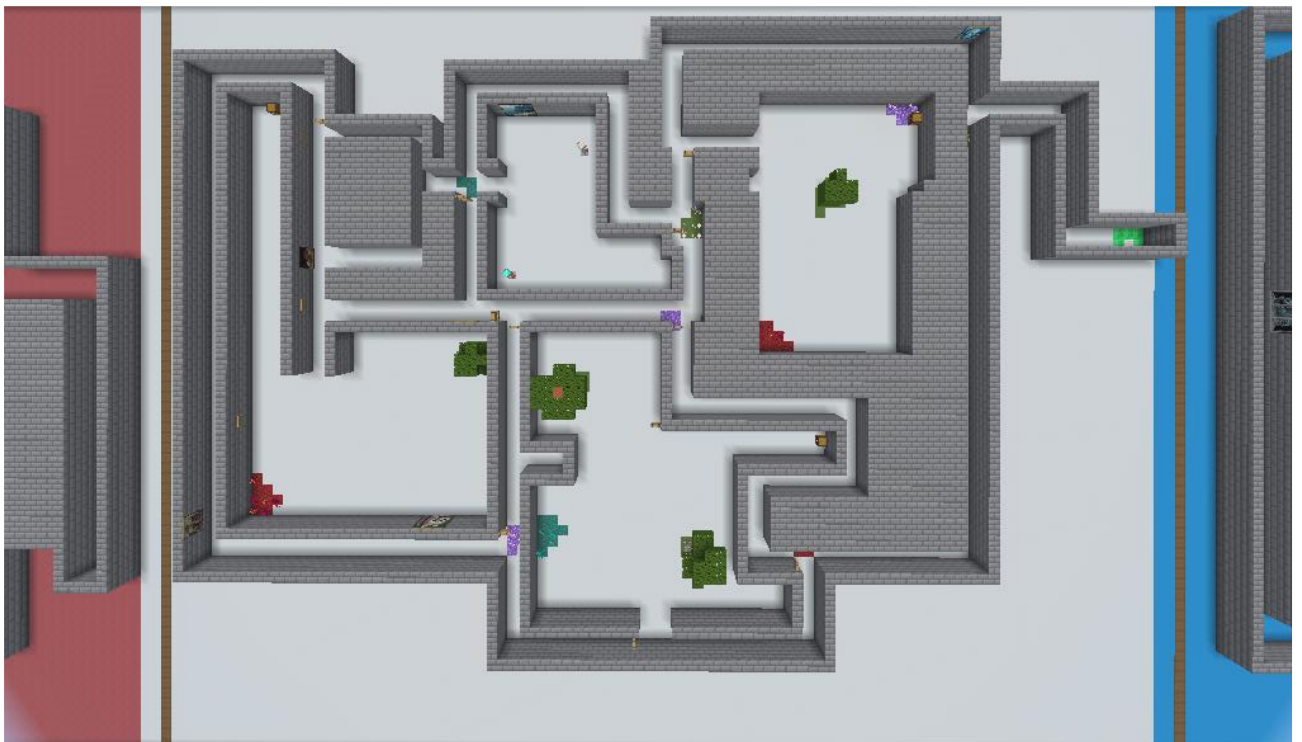


Figure 4: An overhead view of the white map used in the experiment.

Appendix B

Technical Implementation: Command Block System Architecture

For the purpose of helping the experimental process and making sure that the experiments can be repeated, there were multiple command block mechanisms designed and used for the map environments. They played the following roles during the experimental process:

Map Initialization Mechanism

Upon participant entry into the map for the experiment, an entry mechanism had the following two major tasks to accomplish: (1) automatically restocking each chest matching the items with the given collection items, ensuring the same starting point for the experiment, and (2) showing the game toast message for the participant's entry into the map.

Environment Reset Mechanism

To make sure that the environment state stays the same for each next participant, an environment reset mechanism was put into place to systematically remove any leftover items or blocks from the earlier participant. This ensured that all participants had the same starting environment, thus removing any confusing factors that could come from the presence of leftover items.

Item Collection Tracking System

One detection system for each item collection chest was installed that would send an alert message every time an individual gathered an item. This interactive system had multiple uses. First, it gave the individual an immediate response that confirmed the collection of items. Second, it helped track the progress of the participants.

Participant Distribution System

To handle the allocation of participants and the observers, there was an overall design for the teleportation system. This design involved the following sequence for the whole process of teleportation, which would start after the instructions for the tasks were shown to the participants. These were (1) the teleportation of the participant who was not an administrator to their given map scenario, and (2) the teleportation of a given observer to the same site, approximately 20 blocks above the participants. This allowed the observer to unobtrusively monitor the participant.